A PERSONAL LEARNING PLAN • PREPARED FOR ANIRUDH • VERSION 2 (ALL LINKS VERIFIED)

The 6-Month AI Engineer Roadmap

Foundations → Machine Learning → Deep Learning → LLMs → RAG & Agents → Deployment
W3Schools-style interactive lessons • the best GitHub curricula • videos • projects • all free

26 weeks

5 days / week

1–2 hours / day

~200 total hours

3 portfolio projects

START HERE ↓

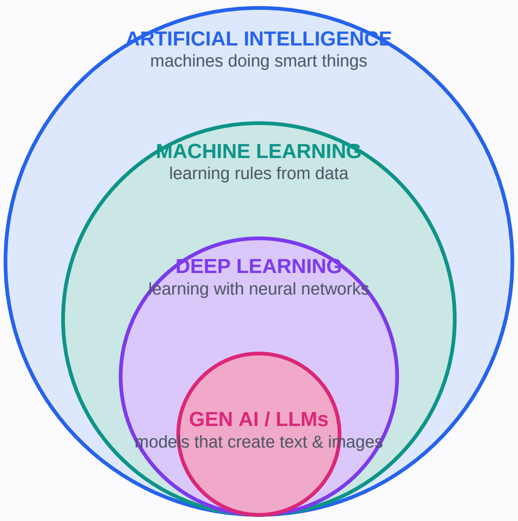
↑ **AI ENGINEER (job-ready)**

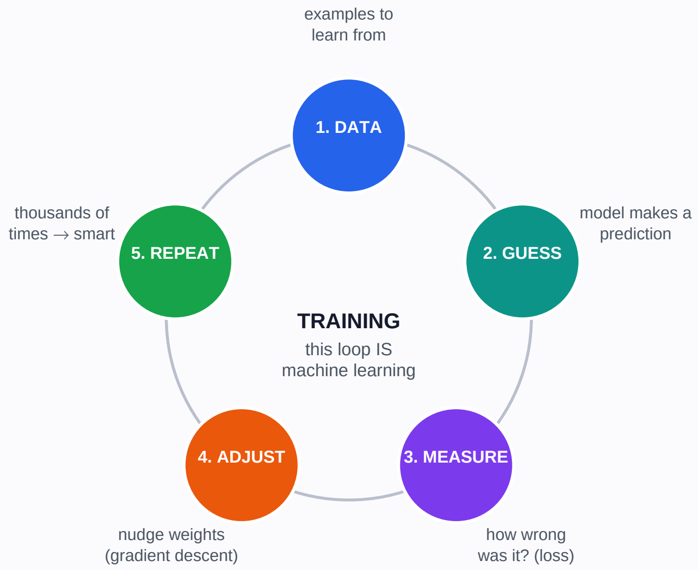
MONTH 1	MONTH 2	MONTH 3	MONTH 4	MONTH 5	MONTH 6
Foundations	**Machine Learning**	**Deep Learning**	**LLMs & Transformers**	**RAG & Agents**	**Deploy & Portfolio**
Python + Math for AI	scikit-learn, first models	Neural nets, PyTorch	GPT, Hugging Face, prompts	Vector DBs, AI apps	Ship 3 projects, interview prep

“You do not need to be great to start. You need to start to become great.” This document is your map. Follow it one week at a time and the destination takes care of itself.

First: What Even Is AI? (Plain-English Basics)

Before any course, get the big picture. People throw around “AI”, “ML”, “deep learning”, and “LLMs” as if they were the same thing. They are not — they are circles inside circles, and the diagram below is the single most important mental model in this whole document.





Your beginner dictionary (refer back here anytime)

Model	The “brain”: a math function with millions of adjustable numbers (weights) that turns input into output.
Training	Showing the model many examples and adjusting its weights until its guesses become good (the loop above).
Dataset	The collection of examples used for training. Data quality matters more than fancy algorithms.
Neural network	A model built from layers of simple units. “Deep learning” just means many layers.
LLM (Large Language Model)	A giant neural network trained on huge amounts of text to predict the next word. GPT and Claude are LLMs.
Token	A piece of a word. LLMs read and write tokens, not letters. “Learning” might be 2 tokens.
Prompt	The instruction you give an LLM. Writing good prompts is a real engineering skill.
Embedding	Turning text into a list of numbers so a computer can measure “meaning similarity” between texts.
RAG	Retrieval-Augmented Generation: letting an LLM look up YOUR documents before answering (diagram in Phase 5).
Agent	An LLM in a loop that can use tools (search, code, APIs) to complete multi-step tasks by itself.
Fine-tuning	Taking a trained model and training it a little more on your own examples to specialize it.
Inference	Using a trained model to get answers. Training = school; inference = the job.

How This Roadmap Works

You said you are scared you will not finish. Remember: you already completed an MS in Computer Science and multiple certifications — those took more discipline than this. The plan needs about 200 hours over 26 weeks. The only failure mode is stopping. Three rules:

- **Consistency beats intensity.** 1–2 focused hours, 5 days a week. Protect the slot like a client meeting.
- **Build more than you watch.** Every “COMPLETE THIS” column item is mandatory. Watching feels like progress; building IS progress.
- **Done beats perfect.** 80% understanding + a working project beats 100% theory with nothing built. If stuck 2 days, mark it and move on.

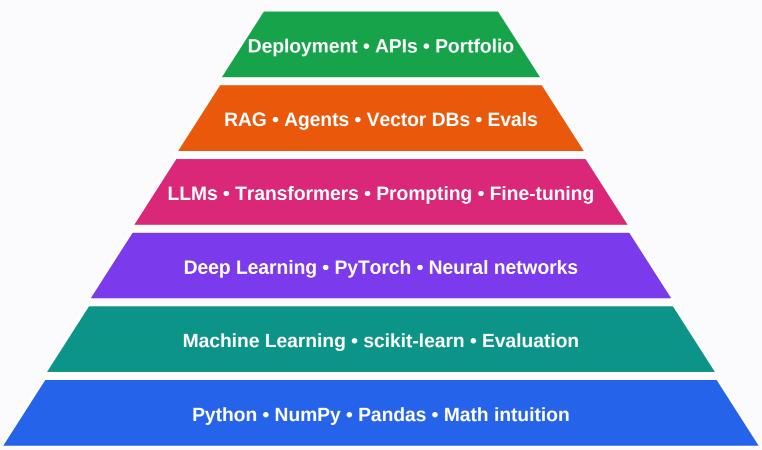
YOUR WEEKLY RHYTHM — 1 to 2 hours per day, 5 days a week (weekends fully off)

MON	TUE	WED	THU	FRI
Watch / read	**Code along**	**Watch / read**	**Code along**	**Build & review**
New concept (videos, course)	Reproduce what you watched	Next concept + notes	Exercises, Kaggle practice	Mini-project + weekly recap

Every phase page gives you 4 things

- **A week-by-week table** — what to learn, the exact link, and a concrete ‘complete this’ checklist item.
- **An EASY MODE box** — W3Schools-style interactive lessons if a topic feels heavy. Same concepts, gentler slope.
- **A MILESTONE PROJECT** — what you push to GitHub at month end. Your GitHub becomes your resume.
- **A checkpoint box** — questions to pass before moving on. If you cannot answer, review; do not skip.

Each layer builds on the one below — do not skip levels



Why this exact order?

AI is a stack, not a menu. Python and math intuition let you understand ML. ML concepts (training, loss, overfitting, evaluation) carry straight into deep learning. Deep learning is what LLMs like GPT and Claude are made of. And RAG, agents, and deployment are how companies actually *use* LLMs — which is precisely the job of an **AI Engineer / Forward Deployed Engineer**.

Skipping layers is why most self-taught journeys fail: things stop making sense. This roadmap never skips, and it never lingers longer than needed.

Cross-check anytime: the community skill-tree at roadmap.sh/ai-engineer — an interactive map where you can tick off topics as you go.

1 PHASE 1: Foundations — Python & Math for AI

Weeks 1–4 • ~30–35 hours total

MONTH 1

Weeks 1–4 of 26 Week 1

Wk 1 Wk 26

Goal: write Python confidently for data work (NumPy, Pandas, charts) and build *visual intuition* for the math behind AI. No textbook proofs — you watch beautiful animations, then immediately use the idea in code. With your CS background, Weeks 1–2 are a fast refresher.

WK	WHAT YOU LEARN (in plain words)	WHERE — exact free resource	COMPLETE THIS ✓
1	Python for data work: functions, classes, list comprehensions, files, Jupyter notebooks.	**Kaggle Learn – Python (interactive, runs in browser)** kaggle.com/learn/python	■ All 7 Kaggle Python lessons + exercises ■ VS Code + Jupyter installed ■ GitHub repo "ai-journey" created
2	NumPy arrays (the language of AI data) and Pandas DataFrames: load, filter, group, join; Matplotlib charts.	**Kaggle Learn – Pandas** kaggle.com/learn/pandas **W3Schools – NumPy tutorial** w3schools.com/python/numpy	■ All 6 Kaggle Pandas lessons ■ W3Schools NumPy: Intro → Array Slicing sections ■ Analyze 1 real CSV: 10 questions, 5 charts
3	Linear algebra visually: vectors, matrices, dot products, transformations — how neural networks "think".	**3Blue1Brown – Essence of Linear Algebra** 3blue1brown.com/topics/linear-algebra (also on YouTube: search "3Blue1Brown linear algebra")	■ Videos 1–7 of the series ■ Recreate dot product + matrix transform in NumPy ■ 1-page notes in your own words
4	Calculus intuition (derivatives = slopes = how models learn) + statistics essentials (mean, variance, distributions, probability).	**3Blue1Brown – Essence of Calculus (videos 1–4)** 3blue1brown.com/topics/calculus **StatQuest – Statistics Fundamentals playlist** youtube.com/@statquest	■ 4 calculus videos + 6 StatQuest videos ■ Code gradient descent from scratch in NumPy to fit a line (20 lines)

EASY MODE (W3Schools-style, interactive in your browser)

If videos feel too fast, do these first — short pages with "Try it Yourself" editors, exactly like learning HTML on W3Schools:

■ **W3Schools Python:** w3schools.com/python — sections "Python Intro" through "Python Classes"
■ **W3Schools NumPy:** w3schools.com/python/numpy — through "Array Reshape"
■ **W3Schools Pandas:** w3schools.com/python/pandas — all sections
■ **W3Schools Matplotlib:** w3schools.com/python/matplotlib_intro.asp — through "Pie Charts"

MILESTONE PROJECT — Data Explorer Notebook

Pick any Kaggle dataset you find fun (stocks, real estate, movies). One polished notebook: load → clean → explore → visualize → 5 insights in plain English. Push to your **ai-journey** repo. This repo grows for all 6 months.

Checkpoint — ready for Phase 2 when you can:

■ Write a Python function and class without looking anything up
■ Explain what matrix multiplication does to a vector (rotate/stretch space)
■ Answer questions about a CSV using Pandas groupby and filtering
■ Explain gradient descent in one sentence: "follow the slope downhill to reduce error"

2 PHASE 2: Machine Learning Fundamentals

Weeks 5–8 • ~32–36 hours total

MONTH 2

Weeks 5–8 of 26

Week 1

Wk 1

Wk 26

Goal: understand how machines learn from data, train real models with scikit-learn, and learn to *evaluate* them honestly — the skill that separates engineers from tutorial-followers. Your structured backbone this month is **Microsoft's ML-For-Beginners**: a famous free GitHub curriculum (26 lessons with quizzes) — exactly the "clear GitHub path" you asked about — plus Andrew Ng's legendary course for depth.

WK	WHAT YOU LEARN (in plain words)	WHERE — exact free resource	COMPLETE THIS ✓
5	What ML is; regression (predicting numbers); classification (predicting categories); cost functions.	**Microsoft ML-For-Beginners (GitHub, 26 lessons)** github.com/microsoft/ML-For-Beginners **Google ML Crash Course (interactive)** developers.google.com/machine-learning/crash-course	■ ML-For-Beginners: Intro (lessons 1–4) + Regression (5–8) ■ Train linear + logistic regression in scikit-learn
6	Decision trees, random forests, gradient boosting — the workhorses of real-world tabular ML.	**Kaggle – Intro to ML (7 lessons)** kaggle.com/learn/intro-to-machine-learning **Kaggle – Intermediate ML (7 lessons)** kaggle.com/learn/intermediate-machine-learning	■ Both Kaggle courses, all exercises ■ Enter Kaggle Titanic competition, submit a real score
7	Evaluation: train/test splits, cross-validation, overfitting vs underfitting, accuracy vs precision/recall/F1, ROC.	**StatQuest – ML playlist (confusion matrix, ROC, bias/variance)** youtube.com/@statquest **scikit-learn User Guide – model evaluation** scikit-learn.org/stable/user_guide.html	■ 8 StatQuest ML videos ■ Improve your Titanic score with cross-validation + feature engineering
8	Clustering (k-means) and dimensionality reduction (PCA); the full end-to-end ML workflow.	**ML-For-Beginners: Classification + Clustering sections** github.com/microsoft/ML-For-Beginners **Andrew Ng ML Specialization — click Enroll → "Audit" = free** coursera.org/specializations/machine-learning-introduction	■ ML-For-Beginners clustering lessons + quiz ■ Full pipeline on Kaggle House Prices: EDA → features → model → evaluate

EASY MODE (W3Schools-style, interactive in your browser)
Gentler on-ramp for this month's concepts:
■ **W3Schools Machine Learning:** w3schools.com/python/python_ml_getting_started.asp — all ML sections (mean/median → decision trees)
■ **W3Schools AI:** w3schools.com/ai — "ML Intro" through "ML Terminology" for friendly definitions with visuals

MILESTONE PROJECT — End-to-End ML Project

Choose a dataset matching your interests (real-estate prices fits your architecture/investing side). Deliver: cleaned data, 2+ models compared, honest cross-validated evaluation, a README explaining every choice. Push to GitHub; post the story on LinkedIn — it builds your AI brand early.

Checkpoint — ready for Phase 3 when you can:
- ■ Explain overfitting and two ways to fight it (more data, regularization, simpler model)
- ■ Choose precision vs recall for a business case and justify it
- ■ Train, tune, and compare two models in scikit-learn without a tutorial
- ■ Score in the top half of the Titanic leaderboard

3 PHASE 3: Deep Learning & PyTorch

Weeks 9–13 • ~40–45 hours total

MONTH 3

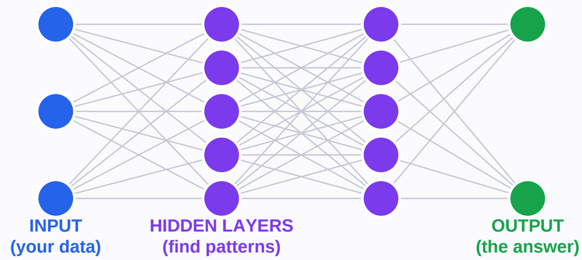
Weeks 9–13 of 26 Week 1

Wk 1 Wk 26

Goal: understand neural networks deeply enough to build one from scratch, then use PyTorch (the industry framework) fluently. Every LLM is a neural network like the one on the right — just enormously bigger.

Your guide: **Andrej Karpathy's "Neural Networks: Zero to Hero"**, free lectures by an OpenAI founding member, widely considered the best deep-learning material ever made. Rule for the month: **type every line yourself. Never copy-paste.**

A neural network: layers of simple math units. Deep = many layers. This is what you build in Phase 3.



WK	WHAT YOU LEARN (in plain words)	WHERE — exact free resource	COMPLETE THIS ✓
9	What a neural network IS: neurons, layers, weights, activations — all in animation first.	**3Blue1Brown – Neural Networks series** 3blue1brown.com/topics/neural-networks **Microsoft AI-For-Beginners (GitHub) – Neural Networks section** github.com/microsoft/AI-For-Beginners	■ All 4 core 3B1B videos, twice ■ Trace one forward pass by hand on paper
10	Backpropagation from scratch: build "micrograd", a tiny 100-line learning engine, line by line.	**Karpathy – Zero to Hero, video 1 (micrograd, 2.5 hrs)** karpathy.ai/zero-to-hero.html	■ Full video, typing along (split over 3 days) ■ Do the exercises in the video description
11	PyTorch: tensors, autograd, nn.Module, the training loop; free GPUs on Google Colab.	**learnpytorch.io – Zero to Mastery PyTorch (free book+videos)** learnpytorch.io **Official PyTorch tutorials** docs.pytorch.org/tutorials	■ learnpytorch.io chapters 00–03 ■ Train an MNIST digit classifier in Colab
12	Computer vision: CNNs and transfer learning — reuse a pretrained model for your own task.	**fast.ai – Practical Deep Learning, lessons 1–2** course.fast.ai	■ Both lessons + notebooks ■ Fine-tune a pretrained model on images you choose (e.g., building styles)
13	Language begins: tokens, embeddings, character-level language models — why attention was invented.	**Karpathy – Zero to Hero, videos 2–3 (makemore)** karpathy.ai/zero-to-hero.html	■ Both videos, typing along ■ Train makemore on names; generate new names — your first language model!

EASY MODE (W3Schools-style, interactive in your browser)
■ **W3Schools AI – Neural Networks:** w3schools.com/ai/ai_neural_networks.asp (Perceptrons → Deep Learning) ■ **TensorFlow Playground:** playground.tensorflow.org — drag sliders, watch a neural net learn live (30 fun minutes)

MILESTONE PROJECT — Image Classifier + Tiny Language Model

To GitHub: (1) fine-tuned image classifier with a demo GIF in the README, (2) your character-level language model with sample generations. You can now truthfully say: "I built neural networks from scratch."

Checkpoint — ready for Phase 4 when you can: ■ tell the backprop story (chain rule + follow the gradient downhill) ■ write a PyTorch training loop from memory ■ explain embeddings ■ explain transfer learning

4 PHASE 4: LLMs & Transformers

Weeks 14–17 • ~32–36 hours total

MONTH 4

Weeks 14–17 of 26 Week 1

Wk 1 Wk 26

Goal: understand how GPT-class models actually work (you will build one!), then get fluent with the modern LLM toolbox: Hugging Face, prompting, structured outputs, and tool calling. Two GitHub curricula anchor this month: **Microsoft's generative-ai-for-beginners** (21 lessons) and **mlabonne's llm-course** (the most-starred LLM roadmap on GitHub).

WK	WHAT YOU LEARN (in plain words)	WHERE — exact free resource	COMPLETE THIS ✓
14	The Transformer and “attention” — the architecture inside every modern LLM. Explore it in 3D first.	**LLM Visualization – walk through a live 3D GPT** bbycroft.net/llm **The Illustrated Transformer (visual essay)** jalammar.github.io/illustrated-transformer	■ Full 3D walkthrough ■ Read essay twice ■ Write a 1-page “how attention works” in your own words
15	Build GPT from scratch with Karpathy: a small transformer, trained end to end, generating text.	**Karpathy – “Let's build GPT” (2 hrs) + “GPT Tokenizer”** karpathy.ai/zero-to-hero.html	■ Train your mini-GPT on Shakespeare in Colab ■ Commit annotated code + sample outputs
16	The open-source ecosystem: Hugging Face transformers, pipelines, model hub; run models locally with Ollama.	**Hugging Face – LLM Course (free, hands-on)** huggingface.co/learn **Ollama – run Llama/Mistral on your laptop** ollama.com	■ HF course chapters 1–2 ■ 3 pipeline tasks (sentiment, summarize, generate) ■ Chat with a local model via Ollama
17	Frontier APIs: system prompts, few-shot prompting, structured JSON output, tool/function calling; fine-tuning & LoRA (concepts only).	**Microsoft generative-ai-for-beginners – lessons 1–6** github.com/microsoft/generative-ai-for-beginners **Anthropic Courses – prompt engineering notebooks** github.com/anthropics/courses	■ 6 Microsoft lessons + quizzes ■ Build a CLI assistant that uses tool calling to fetch live data

EASY MODE (W3Schools-style, interactive in your browser)
Gentle intros if transformers feel intimidating:
■ **W3Schools Generative AI:** w3schools.com/gen_ai — short friendly pages on LLMs, prompts, and chatbots
■ **poloclub.github.io/transformer-explainer** — another interactive in-browser transformer you can poke at

MILESTONE PROJECT — “I Built GPT” + LLM Toolkit

Publish your trained mini-GPT with sample outputs, plus the tool-calling CLI assistant. Write the LinkedIn post: “I built a GPT from scratch — here's what I learned about how LLMs really work.” Recruiters notice posts like this.

Checkpoint — ready for Phase 5 when you can:
- ■ Explain attention in plain English: each word looks at other words and decides what matters
- ■ Explain tokens, context windows, and temperature
- ■ Get reliable JSON out of an LLM API with a system prompt
- ■ State the difference between pretraining, fine-tuning, and prompting

5 PHASE 5: RAG, Agents & AI Engineering

Weeks 18–22 • ~40–45 hours total

MONTH 5

Weeks 18–22 of 26 Week 1

Wk 1 Wk 26

Goal: master the patterns companies hire for — RAG (LLMs that answer from your documents), agents (LLMs that take actions), and evaluation — it maps one-to-one onto AI Engineer and FDE job descriptions. Study this diagram until you can redraw it from memory:

HOW RAG WORKS — "Retrieval-Augmented Generation" = give the AI your documents so it answers from facts, not memory

YOUR DOCS → SPLIT INTO CHUNKS → EMBED (→ numbers) → VECTOR DATABASE → RETRIEVE TOP MATCHES → LLM WRITES ANSWER

User asks a question → the question is embedded too → the database finds the most similar chunks → those chunks + the question go to the LLM → grounded answer with citations

WK	WHAT YOU LEARN (in plain words)	WHERE — exact free resource	COMPLETE THIS ✓
18	Embeddings & vector search: chunking documents, similarity search, vector databases (Chroma; pgvector pairs with your Supabase experience).	**DeepLearning.AI short courses – "Embeddings" + "Vector Databases" (free)** deeplearning.ai/short-courses	■ 2 short courses ■ Embed 50 docs into Chroma; run 10 semantic searches and inspect results
19	RAG end-to-end: retrieve → augment → generate; grounding with citations; where RAG breaks.	**DeepLearning.AI – LangChain "Chat with Your Data"** deeplearning.ai/short-courses **LangChain docs (or LlamaIndex — pick one)** python.langchain.com	■ Course + docs quickstart ■ Build "chat with my PDFs" over documents you know well
20	Improving RAG: hybrid search, reranking, metadata filters; measuring quality (faithfulness, relevance).	**Ragas – RAG evaluation framework (GitHub)** github.com/explodinggradients/ragas	■ Add reranking + a 10-question eval set ■ Measure and record before/after scores
21	Agents: reasoning loops, tool use, memory, multi-step workflows; Model Context Protocol (MCP) basics.	**Microsoft gen-ai-for-beginners – agent + RAG lessons** github.com/microsoft/generative-ai-for-beginners **MCP official docs** modelcontextprotocol.io	■ Agent lessons + quizzes ■ Build an agent that plans, calls 2+ tools (search, calculator, your RAG), reports
22	Engineering practice: streaming, caching, cost/latency tradeoffs, guardrails, evals as regression tests.	**Hamel Husain – "Your AI product needs evals" (on hamel.dev blog)** hamel.dev	■ Read the essay, take notes ■ Add streaming, error handling, and the eval suite to your app

MILESTONE PROJECT — Production-Style RAG + Agent App (your flagship)

One serious application: document Q&A; with citations, an agent mode with tools, an eval suite, and a clean UI — you already know Next.js, use it. This is your #1 interview project, and the exact skillset feeds reknitAI's AI features.

Checkpoint — ready for Phase 6 when you can:
- ■ Redraw the RAG pipeline above from memory in 2 minutes
- ■ Name 3 reasons a RAG system answers wrongly, with a fix for each
- ■ Explain what makes something an "agent" vs a chatbot (loop + tools + goal)
- ■ Describe how you would evaluate an LLM app before shipping

6 PHASE 6: Deploy, Portfolio & Interview Prep

Weeks 23–26 • ~30–34 hours total

MONTH 6

Weeks 23–26 of 26 Week 1

Wk 1 Wk 26

Goal: turn skills into evidence. Ship projects publicly, polish GitHub and resume, and prepare for AI Engineer / Forward Deployed Engineer interviews. FDE roles reward exactly your profile: consulting experience + client communication + hands-on AI building.

WK	WHAT YOU LEARN (in plain words)	WHERE — exact free resource	COMPLETE THIS ✓
23	Serving AI apps: FastAPI endpoints, Docker basics, config and secrets management.	**FastAPI official tutorial (excellent)** fastapi.tiangolo.com **Docker – Get Started guide** docker.com/get-started	■ FastAPI tutorial parts 1–10 ■ Wrap your RAG app in an API; containerize it
24	Deploying: host your API + frontend publicly; logging, monitoring, and cost awareness.	**Hugging Face Spaces – free demo hosting** huggingface.co/spaces **Vercel (you already use it) for the frontend** vercel.com	■ Public URL live for your flagship app ■ Add the URL to your resume
25	Portfolio polish: READMEs with architecture diagrams, demo GIFs; write 2 technical posts.	**readme.so – README builder** readme.so **Excalidraw – architecture diagrams** excalidraw.com	■ 3 pinned repos, each with diagram + demo + writeup ■ 2 posts published (LinkedIn or dev.to)
26	Interviews: ML fundamentals Q&A;, LLM system design (“design a RAG system”), behavioral stories; apply.	**ML Interviews study guide (GitHub)** github.com/alirezadir/Machine-Learning-Interviews **deep-ml.com – ML coding practice problems** deep-ml.com	■ 20 practice questions answered aloud ■ 2 mock interviews (use Claude!) ■ 10 tailored applications sent

MILESTONE PROJECT — Graduation: The Job-Ready Package

By end of Week 26: 3 deployed projects with public URLs, a GitHub that tells a 6-month story, 2 technical posts, an AI-focused resume, and practiced interview answers. That is not “learning AI” — that is being an AI engineer.

The GitHub Learning Paths (star these today)

You mentioned seeing GitHub repos with clear AI paths — these are the best ones, and the roadmap above already weaves them into your weeks. Star them all now so they are one click away. Column 3 tells you exactly which parts to complete.

microsoft / ML-For-Beginners github.com/microsoft/ML-For-Beginners	Microsoft's famous free curriculum: 26 lessons on classic ML with quizzes, sketchnotes, and assignments.	Intro + Regression + Classification + Clustering sections → **Phase 2 (Weeks 5–8)**
microsoft / AI-For-Beginners github.com/microsoft/AI-For-Beginners	24-lesson curriculum covering neural networks, computer vision, and NLP with runnable notebooks.	Neural Networks + Computer Vision sections → **Phase 3 (Weeks 9–12)**
karpathy / nn-zero-to-hero github.com/karpathy/nn-zero-to-hero	Code companion to the Zero to Hero videos: micrograd, makemore, and build-GPT notebooks.	micrograd + makemore → **Weeks 10, 13**; build-GPT → **Week 15**
microsoft / generative-ai-for-beginners github.com/microsoft/generative-ai-for-beginners	21 lessons on building GenAI apps: prompting, RAG, agents, fine-tuning, security — with videos and quizzes.	Lessons 1–6 → **Week 17**; RAG + agent lessons → **Weeks 19–21**
mlabonne / llm-course github.com/mlabonne/llm-course	The most-starred LLM roadmap on GitHub: three tracks (Fundamentals, LLM Scientist, LLM Engineer) with curated links.	Use as a map, not a course: skim the LLM Engineer track in **Phase 5** and fill any gaps you notice
rasbt / LLMs-from-scratch github.com/rasbt/LLMs-from-scratch	Build a ChatGPT-style model step by step in PyTorch (companion to Sebastian Raschka's book code, free on GitHub).	Optional deep-dive after **Week 15** if you loved building GPT
roadmap.sh AI Engineer roadmap.sh/ai-engineer	Interactive community skill-tree where you tick off topics and open linked resources per node.	Open monthly; mark completed nodes — great motivation and gap-checking
armankhondker / awesome-ai-ml-resources github.com/armankhondker/awesome-ai-ml-resources	A curated roadmap + free resource list, beginner to advanced.	Reference only — when you want an alternative explanation of any topic

Interactive websites that make concepts click (W3Schools-style)

W3Schools Python / NumPy / Pandas / ML / AI w3schools.com/python • w3schools.com/ai	Short pages + “Try it Yourself” editors. The exact style you asked for.	Easy Mode boxes in **Phases 1–4** list the exact sections
Kaggle Learn kaggle.com/learn	Free micro-courses where code runs in the browser and is auto-checked.	Python, Pandas → **Phase 1**; Intro + Intermediate ML → **Phase 2**
TensorFlow Playground playground.tensorflow.org	Drag sliders, watch a neural network learn live in your browser.	30 minutes in **Week 9** — pure intuition-building
LLM Visualization bbycroft.net/llm	A 3D animated GPT you can walk through, operation by operation.	Full walkthrough in **Week 14**
Google ML Crash Course developers.google.com/machine-learning/crash-course	Google's free interactive course with visualizations and exercises.	Alongside **Weeks 5–7** as a second explanation of anything unclear

Your Fears, Answered Honestly

“Will I really be able to complete this?”

Yes — if you protect the daily slot. ~200 hours over 26 weeks, less than your MS took. The only failure mode is stopping. Miss a week? Resume where you left off. Never restart from Week 1.

“Will I be an expert in 6 months?”

Honest answer: you will be a **competent, hireable AI engineer** — able to build, evaluate, and deploy real AI systems and speak about them credibly. “Answer anything” expertise takes years for everyone. Six months gets you to the level where you learn the rest on the job — which is how every expert actually got there.

“What if I get stuck?”

Rule of three: (1) re-watch at 0.75x speed, (2) ask Claude to explain with an analogy and quiz you, (3) still stuck after 2 days → mark it, move on. Later material often makes earlier material click.

“Do I need money or a GPU?”

No. Colab gives free GPUs; every course here is free or free-to-audit (on Coursera click Enroll → “Audit”). Optional: $5–10 of API credits in Phases 4–5; free tiers usually cover it.

“Is math going to block me?”

You need intuition, not proofs. 3Blue1Brown gives you pictures; the code gives you practice. If you can follow “slope tells you which way is downhill,” you can follow deep learning.

“What exactly is a Forward Deployed Engineer?”

An engineer who sits with customers and builds AI solutions on their real data and workflows — half builder, half consultant. Your ISI/Deloitte consulting background plus this roadmap is nearly a perfect FDE profile.

Your First Day Starts Now

Do not wait for Monday. Your first 60 minutes, today:

- **Minute 0–10:** Create the GitHub repo **ai-journey**; paste the 6 phases into its README. Star the repos from the GitHub directory page.
- **Minute 10–20:** Open **kaggle.com/learn/python**, create your account, start Lesson 1.
- **Minute 20–55:** Finish Lesson 1 exercises. Actually type the code.
- **Minute 55–60:** Put a recurring 1-hour block on your calendar, Mon–Fri. Unmissable.

MILESTONE PROJECT — The Promise of Week 26

Twenty-six weeks from now: you understand LLMs because you built one. You shipped a RAG application with an agent and evals. Your GitHub shows a 6-month arc from Pandas to production AI. You can walk into an AI Engineer or Forward Deployed Engineer interview and talk about systems you actually built. The person who finishes this roadmap is not the person who started it. Begin.

Weekly habit tracker (print this page)

Wk	M	T	W	T	F	Built something?	Note to self
1							
2							
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When you finish a phase, come back and tell me — I will quiz you on the checkpoints, help debug your projects, and adjust the plan. You are not doing this alone.